Module 1, Chapter 2: Engineering Risk & Operations

The Language of the Underwriter

- **Risk:** The fundamental uncertainty concerning a loss, serving as the actual subject matter of an insurance policy.
  - **Speculative Risk:** Can result in either a gain or a loss (e.g., betting, stocks) and is **entirely uninsurable**.
  - **Pure Risk:** Can produce a loss only, with absolutely zero chance of financial gain. This is the only type of risk insurance handles.
- **Peril:** The actual, immediate cause of the loss—the unpredictable event insured against, such as a fire, earthquake, or shipwreck.
- **Hazard:** Factors that actively increase the chance of loss from a peril. Underwriters classify them into three buckets:
  - **Physical Hazards:** Tangible, structural dangers you can see (e.g., stockpiling combustible goods, exposed wiring, defective tail lights).
  - **Moral Hazards:** Internal vulnerabilities related to the character, dishonesty, or poor credit standing of the insured.
  - **Morale Hazards:** Factors rooted in a person's negative attitude, carelessness, or indifference to a loss precisely because they have insurance coverage (e.g., poor housekeeping).

Financial Metrics & Strict Rules

- **Total Sum Insured (TSI):** The "face amount" of the policy representing the absolute maximum limit of the Insurer's legal liability.
- **Premium:** The consideration value paid by the insured for the assumption of loss.
  - **Strict Rule:** No Premium = No Policy and No Liability.
  - **Legal Exceptions:** Documented credit terms, or an established grace period in Life Insurance.
- **Gross Premiums Written (GPW):** The total premium production of an insurer for a specific period.
- **Reinsurance:** Defined as "an insurance of an insurance," where a primary insurer shares a portion of their total sum insured with a professional reinsurer.
- **Net Premiums Written (NPW):** The total premium production minus the costs of Reinsurance.

Operational Gatekeepers

- **Underwriter:** Evaluates, accepts risks, sets terms, and determines the exact liability to retain. Must be licensed by the Insurance Commission.
- **Agent:** Explicitly represents the insurance company. Non-life agents can represent up to 7 companies; life agents can only represent 1.
- **Broker:** Explicitly represents the client. Must be incorporated, registered with the SEC, and utilize a certified Soliciting Official.

Criteria for Insurable Risks

- Must have a large number of homogeneous exposure units utilizing the **Law of Large Numbers**.
- Simultaneous or catastrophic losses must be minimal.
- The event must be definite and measurable.
- The cost must be economically feasible with a calculable chance of loss.

> **Key Takeaway:** Master the flow of terminology. The *Peril* causes the loss, the *Hazard* makes it more likely to happen, and the *Risk* is the fundamental uncertainty of it all. Remember: Agents represent the company, Brokers represent the client!